

Problem

Solve the following differential equations subject to the specified initial conditions.

(a) $d^2v/dt^2 + 4v = 12$, $v(0) = 0$, $dv(0)/dt = 2$

(b) $d^2i/dt^2 + 5di/dt + 4i = 8$, $i(0) = -1$, $di(0)/dt = 0$

(c) $d^2v/dt^2 + 2dv/dt + v = 3$, $v(0) = 5$, $dv(0)/dt = 1$

(d) $d^2i/dt^2 + 2di/dt + 5i = 10$, $i(0) = 4$, $di(0)/dt = -2$

Step-by-step solution

Step 1 of 12

(a)

Consider the following differential equation:

$$\frac{d^2v}{dt^2} + 4v = 12$$

Consider that the differential equation subject to the following initial conditions:

$$v(0) = 0$$

$$v'(0) = 2$$

Take the Laplace transform of the each term in the differential equation:

$$\left[s^2V - sv(0) - v'(0) \right] + 4V = \frac{12}{s}$$

Substitute the initial conditions.

$$\left[s^2V - s(0) - 2 \right] + 4V = \frac{12}{s}$$

$$s^2V + 4V = \frac{12}{s} + 2$$

$$(s^2 + 4)V = \frac{2s + 12}{s}$$

$$V = \frac{2s + 12}{s(s^2 + 4)}$$

Step 2 of 12

Use partial fraction method to simplify the function further.

$$\frac{2s+12}{s(s^2+4)} = \frac{A}{s} + \frac{Bs+C}{s^2+4}$$

$$\begin{aligned} 2s+12 &= A(s^2+4) + (Bs+C)s \\ &= As^2 + 4A + Bs^2 + Cs \\ &= (A+B)s^2 + Cs + 4A \end{aligned}$$

Compare the coefficients of the terms to get the following values:

$$C = 2$$

$$4A = 12$$

$$\begin{aligned} A &= \frac{12}{4} \\ &= 3 \end{aligned}$$

$$A+B=0$$

$$3+B=0$$

$$B = -3$$

Write the simplified expression for V .

$$\begin{aligned} V &= \frac{3}{s} + \frac{-3s+2}{s^2+4} \\ &= \frac{3}{s} - \frac{3s}{s^2+4} + \frac{2}{s^2+4} \end{aligned}$$

Step 3 of 12

Find the inverse Laplace transform to solve the differential equation.

Use the following properties of the Laplace transform:

$$u(t) \leftrightarrow \frac{1}{s}$$

$$\sin \omega t \leftrightarrow \frac{\omega}{s^2 + \omega^2}$$

$$\cos \omega t \leftrightarrow \frac{s}{s^2 + \omega^2}$$

Find the inverse Laplace transform of the function V .

$$\begin{aligned} v &= 3u(t) - 3 \cos 2t + \sin 2t \\ &= 3u(t) + 3.162[-0.9487 \cos 2t + 0.316 \sin 2t] \\ &= 3u(t) + 3.162[\cos 2t \cos(161.5^\circ) + \sin 2t \sin(161.5^\circ)] \\ &= [3 + 3.162 \cos(2t - 161.5^\circ)]u(t) \end{aligned}$$

Thus, the solution of the differential equation is $v = [3 + 3.162 \cos(2t - 161.5^\circ)]u(t) \text{ V}$.

Step 4 of 12

(b)

Consider the following differential equation:

$$\frac{d^2i}{dt^2} + \frac{5di}{dt} + 4i = 8$$

Consider that the differential equation subject to the following initial conditions:

$$i(0) = -1$$

$$i'(0) = 0$$

Take the Laplace transform of the each term in the differential equation:

$$\left[s^2 I - si(0) - i'(0) \right] + 5 \left[sI - i(0) \right] + 4I = \frac{8}{s}$$

Substitute the initial conditions.

$$\left[s^2 I - s(-1) - 0 \right] + 5 \left[sI - (-1) \right] + 4I = \frac{8}{s}$$

$$s^2 I + 5sI + 4I = \frac{8}{s} - s - 5$$

$$(s^2 + 5s + 4)I = \frac{8 - s^2 - 5s}{s}$$

$$I = \frac{-s^2 - 5s + 8}{s(s^2 + 5s + 4)}$$

$$= -\frac{s^2 + 5s - 8}{s(s^2 + 4s + s + 4)}$$

$$= -\frac{s^2 + 5s - 8}{s(s + 4)(s + 1)}$$

Use partial fraction method to simplify the function further.

$$\frac{s^2 + 5s - 8}{s(s+4)(s+1)} = \frac{A}{s} + \frac{B}{s+4} + \frac{C}{s+1}$$

$$\begin{aligned} s^2 + 5s - 8 &= A(s^2 + 5s + 4) + B(s^2 + s) + C(s^2 + 4s) \\ &= As^2 + 5As + 4A + Bs^2 + Bs + Cs^2 + 4Cs \\ &= (A + B + C)s^2 + (5A + B + 4C)s + 4A \end{aligned}$$

Compare the coefficients of the terms to get the following values:

$$4A = -8$$

$$A = -2$$

$$A + B + C = 1$$

$$-2 + B + C = 1$$

$$B + C = 3 \dots\dots (1)$$

$$5A + B + 4C = 5$$

$$5(-2) + B + 4C = 5$$

$$B + 4C = 15 \dots\dots (2)$$

Solve equations (1) and (2).

$$B + 4C = 15$$

$$\begin{array}{r} (-) B + C = (-) 3 \\ \hline 3C = 12 \end{array}$$

$$C = 4$$

$$B + C = 3$$

$$B + 4 = 3$$

$$B = -1$$

Write the simplified expression for I .

$$\begin{aligned} I &= -\left[\frac{-2}{s} - \frac{1}{s+4} + \frac{4}{s+1} \right] \\ &= \frac{2}{s} + \frac{1}{s+4} - \frac{4}{s+1} \end{aligned}$$

Step 6 of 12

Find the inverse Laplace transform to solve the differential equation.

Use the following properties of the Laplace transform:

$$u(t) \leftrightarrow \frac{1}{s}$$

$$e^{-at} \leftrightarrow \frac{1}{s+a}$$

Find the inverse Laplace transform of the function I .

$$\begin{aligned} i &= 2u(t) + e^{-4t}u(t) - 4e^{-t}u(t) \\ &= (2 + e^{-4t} - 4e^{-t})u(t) \end{aligned}$$

Thus, the solution of the differential equation is $i = [2 + e^{-4t} - 4e^{-t}]u(t) \text{ A}$.

Step 7 of 12

(c)

Consider the following differential equation:

$$\frac{d^2v}{dt^2} + \frac{2dv}{dt} + v = 3$$

Consider that the differential equation subject to the following initial conditions:

$$v(0) = 5$$

$$v'(0) = 1$$

Take the Laplace transform of the each term in the differential equation:

$$[s^2V - sv(0) - v'(0)] + 2[sV - v(0)] + V = \frac{3}{s}$$

Substitute the initial conditions.

$$[s^2V - s(5) - 1] + 2[sV - 5] + V = \frac{3}{s}$$

$$s^2V + 2sV + V = \frac{3}{s} + 5s + 1 + 10$$

$$(s^2 + 2s + 1)V = \frac{3 + 5s^2 + 11s}{s}$$

$$V = \frac{5s^2 + 11s + 3}{s(s^2 + 2s + 1)}$$

$$= \frac{5s^2 + 11s + 3}{s(s^2 + s + s + 1)}$$

$$= \frac{5s^2 + 11s + 3}{s(s+1)^2}$$

Step 8 of 12

Use partial fraction method to simplify the function further.

$$\frac{5s^2 + 11s + 3}{s(s+1)^2} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2}$$

$$\begin{aligned} 5s^2 + 11s + 3 &= A(s+1)^2 + Bs(s+1) + Cs \\ &= As^2 + 2As + A + Bs^2 + Bs + Cs \\ &= (A+B)s^2 + (2A+B+C)s + A \end{aligned}$$

Compare the coefficients of the terms to get the following values:

$$A = 3$$

$$A + B = 5$$

$$3 + B = 5$$

$$B = 2$$

$$2A + B + C = 11$$

$$2(3) + 2 + C = 11$$

$$8 + C = 11$$

$$C = 3$$

Write the simplified expression for V .

$$V = \frac{3}{s} + \frac{2}{s+1} + \frac{3}{(s+1)^2}$$

Step 9 of 12

Find the inverse Laplace transform to solve the differential equation.

Use the following properties of the Laplace transform:

$$u(t) \leftrightarrow \frac{1}{s}$$

$$e^{-at} \leftrightarrow \frac{1}{s+a}$$

$$t^n e^{-at} \leftrightarrow \frac{n!}{(s+a)^{n+1}}$$

Find the inverse Laplace transform of the function V .

$$\begin{aligned} v &= 3u(t) + 2e^{-t}u(t) + 3te^{-t}u(t) \\ &= (3 + 2e^{-t} + 3te^{-t})u(t) \end{aligned}$$

Thus, the solution of the differential equation is $v = [3 + 2e^{-t} + 3te^{-t}]u(t) \text{ V}$.

Step 10 of 12

(d)

Consider the following differential equation:

$$\frac{d^2 i}{dt^2} + \frac{2di}{dt} + 5i = 10$$

Consider that the differential equation subject to the following initial conditions:

$$i(0) = 4$$

$$i'(0) = -2$$

Take the Laplace transform of the each term in the differential equation:

$$[s^2 I - si(0) - i'(0)] + 2[sI - i(0)] + 5I = \frac{10}{s}$$

Substitute the initial conditions.

$$[s^2 I - s(4) - (-2)] + 2[sI - 4] + 5I = \frac{10}{s}$$

$$s^2 I + 2sI + 5I = \frac{10}{s} + 4s - 2 + 8$$

$$(s^2 + 2s + 5)I = \frac{10 + 4s^2 + 6s}{s}$$

$$I = \frac{4s^2 + 6s + 10}{s(s^2 + 2s + 5)}$$

$$= \frac{2(2s^2 + 3s + 5)}{s(s^2 + 2s + 5)}$$

$$= 2 \frac{2s^2 + 3s + 5}{s(s^2 + 2s + 5)}$$

Step 11 of 12

Use partial fraction method to simplify the function further.

$$\frac{2s^2 + 3s + 5}{s(s^2 + 2s + 5)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2s + 5}$$

$$\begin{aligned} 2s^2 + 3s + 5 &= A(s^2 + 2s + 5) + (Bs + C)s \\ &= As^2 + 2As + 5A + Bs^2 + Cs \\ &= (A + B)s^2 + (2A + C)s + 5A \end{aligned}$$

Compare the coefficients of the terms to get the following values:

$$5A = 5$$

$$A = 1$$

$$A + B = 2$$

$$1 + B = 2$$

$$B = 1$$

$$2A + C = 3$$

$$2(1) + C = 3$$

$$C = 1$$

Write the simplified expression for I .

$$\begin{aligned} I &= 2 \left[\frac{1}{s} + \frac{s+1}{s^2 + 2s + 5} \right] \\ &= \frac{2}{s} + \frac{2(s+1)}{s^2 + 2s + 1 + 4} \\ &= \frac{2}{s} + \frac{2(s+1)}{(s+1)^2 + 2^2} \end{aligned}$$

Step 12 of 12

Find the inverse Laplace transform to solve the differential equation.

Use the following properties of the Laplace transform:

$$u(t) \leftrightarrow \frac{1}{s}$$

$$e^{-at} \cos \omega t \leftrightarrow \frac{s+a}{(s+a)^2 + \omega^2}$$

Find the inverse Laplace transform of the function I .

$$\begin{aligned} i &= 2u(t) + 2e^{-t} \cos 2tu(t) \\ &= (2 + 2e^{-t} \cos 2t)u(t) \end{aligned}$$

Thus, the solution of the differential equation is $i = [2 + 2e^{-t} \cos 2t]u(t) \text{ A}$.